The sealed statements of LocalAlgebrasLean

These 21 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

Setting

There are sites Λ, a finite alphabet σ_u at every site u, and a reference configuration s_0 choosing a letter at every site (so every alphabet is nonempty). For a finite set of sites X, a matrix on the letters of X is a complex matrix indexed by the assignments ξ of a letter $\xi_x \in \sigma_x$ to each $x \in X$, that is, an operator on $\bigotimes_{x\in X} \mathbb{C}^{\sigma_x}$. Bounded operators carry the operator norm, and C*-algebras are complex.

1. The finitely supported configurations are the configurations s that differ from the reference configuration s_0 at finitely many sites.
`Sealed.finConfig_def`

2. The Hilbert space H is $\ell^2(F)$, the square-summable complex functions on the set F of finitely supported configurations.
`Sealed.configSpace_def`

The local embeddings

For a finite set of sites X, the local embedding ι_X is a unital $*$-homomorphism from the matrices on the letters of X; this is part of the type of ι_X in the Lean statements. Statement 3 determines ι_X by its matrix elements.

3. For a finite set of sites X, a matrix m on the letters of X and finitely supported configurations s, t, the matrix element $\langle e_s, \iota_X(m) e_t\rangle$ in the standard basis of $\ell^2(F)$ is $m(s|_X, t|_X)$ if s and t agree at every site outside X, and 0 otherwise.
`Sealed.inner_localEmbed`

4. For a finite set of sites X, the local embedding ι_X is injective and $\|\iota_X(m)\| = \|m\|$ for every matrix m on the letters of X, the norm of m being its operator norm on ℓ^2.
`Sealed.localEmbed_injective_and_norm`

5. Compatibility: for finite sets of sites $X \subseteq Y$ and a matrix m on the letters of X, $\iota_Y(m \otimes 1) = \iota_X(m)$, where

$$(m \otimes 1)(\zeta, \zeta') = m(\zeta|_X, \zeta'|_X)$$

if the letters ζ, ζ' on Y agree outside X, and 0 otherwise.
`Sealed.localEmbed_extend`

6. Local commutativity: for disjoint finite sets of sites X and Y, $\iota_X(a)$ and $\iota_Y(b)$ commute for all matrices a on the letters of X and b on the letters of Y.
`Sealed.commute_localEmbed_of_disjoint`

The net of local algebras

Every set of sites O has a $*$-subalgebra $\mathcal{A}(O)$ of the bounded operators on H, closed in norm (statement 7), and the quasi-local algebra is $\mathcal{A}_{ql} = \mathcal{A}(\Lambda)$.

7. For every set of sites O, the algebra $\mathcal{A}(O)$ is the norm closure, in the bounded operators on H, of the set of operators $\iota_X(m)$ with $X \subseteq O$ a finite set of sites and m a matrix on the letters of X.
`Sealed.quasiLocalNet_eq_closure`

8. The quasi-local algebra $\mathcal{A}_{ql}$ is the algebra $\mathcal{A}(\Lambda)$ of the set of all sites.
`Sealed.quasiLocalAlgebra_def`

9. Local commutativity: for disjoint sets of sites O_1 and O_2, every element of $\mathcal{A}(O_1)$ commutes with every element of $\mathcal{A}(O_2)$.
`Sealed.commute_of_mem_quasiLocalNet`

10. Isotony: for sets of sites $O_1 \subseteq O_2$, $\mathcal{A}(O_1) \subseteq \mathcal{A}(O_2)$.
`Sealed.quasiLocalNet_mono`

Symmetries and the reference configuration

11. Covariance: let a group Γ act on the sites, with one finite alphabet Q at every site and a reference configuration s_0 with $s_0(gu) = s_0(u)$. For each $g \in \Gamma$ there is a unitary U on H with $U\iota_X(m)U^* = \iota_{gX}(g_*m)$ for every finite set of sites X, where $gX = \{y : g^{-1}y \in X\}$ and

$$(g_*m)(\zeta, \zeta') = m(\zeta \circ g, \zeta' \circ g),$$

and with $U\mathcal{A}(O)U^* = \mathcal{A}(gO)$ for every set of sites O. The unitary is asserted for each g separately; the statement does not say that $g \mapsto U$ is a representation.
`Sealed.exists_unitary_covariance`

12. Independence of the reference configuration: for any two reference configurations s_0 and s_0' (which may differ at infinitely many sites) there is a $*$-isomorphism $\varphi : \mathcal{A}_{ql} \to \mathcal{A}'_{ql}$ of the quasi-local algebras with $a \in \mathcal{A}(O) \iff \varphi(a) \in \mathcal{A}'(O)$ for every set of sites O and every $a \in \mathcal{A}_{ql}$, and with $\varphi(\iota_X(m)) = \iota'_X(m)$ for every finite set of sites X and matrix m on its letters.
`Sealed.exists_starAlgEquiv_of_ref`

The universal property

13. Universal property: let D be a C*-algebra and (β_X) a family, indexed by the finite sets of sites X, of unital $*$-homomorphisms from the matrices on the letters of X to D, compatible in the sense $\beta_Y(m \otimes 1) = \beta_X(m)$ for $X \subseteq Y$ (with $m \otimes 1$ as in the compatibility of the local embeddings). Then there is exactly one unital $*$-homomorphism $\Phi : \mathcal{A}_{ql} \to D$ with $\Phi(\iota_X(m)) = \beta_X(m)$ for all X and m.
`Sealed.exists_unique_lift`

14. Let D be a C*-algebra other than the zero algebra and (β_X) a compatible family as in the universal property. Then the $*$-homomorphism $\Phi : \mathcal{A}_{ql} \to D$ with $\Phi(\iota_X(m)) = \beta_X(m)$ for all X and m is an isometry.
`Sealed.isometry_lift`

Finite volume

15. If the set of sites Λ is finite, the quasi-local algebra $\mathcal{A}_{ql}$ is the algebra of all bounded operators on H.
`Sealed.quasiLocalAlgebra_eq_top_of_finite`

16. For a finite set of sites O, the algebra $\mathcal{A}(O)$ is the range of the local embedding ι_O.
`Sealed.quasiLocalNet_eq_range_of_finite`

Examples

These examples show that the hypotheses of the main theorems can be met and that the objects are not degenerate.

17. For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the quasi-local algebra is not the zero algebra.
`Sealed.witness_nontrivial_quasiLocalAlgebra`

18. For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the set of finitely supported configurations, which indexes the basis of H, is infinite.
`Sealed.witness_infinite_finConfig`

19. For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the algebra $\mathcal{A}(\{0\})$ of the site 0 contains an element that is not a multiple of the identity.
`Sealed.witness_quasiLocalNet_singleton_not_scalar`

20. For the chain $\mathbb{Z}$ with two letters per site and the reference configuration 0, which is invariant under translations, there is a unitary U on H with $U\mathcal{A}(\{0\})U^* = \mathcal{A}(\{1\})$.
`Sealed.witness_translation_covariant`

21. For the chain $\mathbb{N}$ with two letters per site, the reference configurations 0 and 1 differ at infinitely many sites, and there is a $*$-isomorphism φ of their quasi-local algebras with $a \in \mathcal{A}(O) \iff \varphi(a) \in \mathcal{A}'(O)$ for every set of sites O.
`Sealed.witness_references_differing_everywhere`

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